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Some Causal Effects of an Industrial Policy

Contents

1 Overview

- **Question:** Do place-based investment subsidies raise real activity (jobs, investment, output) in disadvantaged areas, and do they improve productivity?
- **Policy:** UK *Regional Selective Assistance (RSA)*: discretionary grants to firms in "assisted areas". EU State Aid rules cap the *maximum investment subsidy rate* allowed in each area.
- **Treatment intensity:** the *maximum investment subsidy* (Net Grant Equivalent, NGE), i.e., the cap on the subsidy rate (in percent).
- **Identification:** exploit *rule changes* in EU eligibility (notably 2000 map reform) that shift the NGE cap differentially across areas. Build a *policy-rule instrument* from (i) estimated rule weights and (ii) predetermined area characteristics.
- **Main results (area level):** A 10pp higher NGE increases manufacturing employment by about 10% and reduces unemployment. No robust effect on non-manufacturing employment.
- **Margins:** effects mainly come from the *intensive margin* (incumbent expansion), not from more plants (no strong evidence for entry).
- **Heterogeneity:** effects concentrate in *small firms*; large firms receive subsidies but do not expand employment (suggesting inframarginal rents, gaming, or weaker additionality).
- **Firm outcomes:** subsidies raise *investment* and *output*, but do *not* raise *TFP*.
- **Macro/TFP intuition:** a capital-subsidy wedge can raise jobs while leaving (or even lowering) measured productivity through misallocation.

2 Institutional background: what the program is

2.1 RSA in one sentence

RSA is a discretionary grant program (UK) targeted to disadvantaged locations. A firm in an eligible area can apply for a subsidy on investment expenditures, but the EU state-aid regime caps the *maximum allowable subsidy rate* in each area.

2.2 Key object: maximum subsidy rate (NGE)

Let $\mathrm{NGE}_{r,t}$ denote the maximum investment subsidy rate in area r and year t (Net Grant Equivalent).

- $\mathrm{NGE}_{r,t} = 0$ if area r is ineligible.
- Otherwise $\mathrm{NGE}_{r,t}$ takes values in discrete tiers (e.g., 10%, 15%, 20%, 30%, 35% depending on the policy regime).

The **map reform around 2000** changes which areas are eligible and what cap applies, generating quasi-experimental variation.

3 A simple model: how an investment subsidy should work

The paper uses a standard user-cost-of-capital logic and then discusses why effects may be stronger for financially constrained firms.

3.1 User cost of capital with an investment subsidy

Let:

- δ = depreciation rate,
- r = required return / interest rate,
- τ = corporate tax rate,
- θ = present value of depreciation allowances (per unit of investment),
- ϕ = investment subsidy rate (think: $\phi \approx$ NGE for the maximum cap).

A standard expression for the user cost of capital is

$$\rho(\phi) = \delta + \frac{r\,(1 - \phi - \theta\tau)}{1 - \tau}. \tag{1}$$

Comparative statics:

$$\frac{\partial \rho}{\partial \phi} = -\frac{r}{1 - \tau} < 0. \tag{2}$$

So a higher subsidy reduces the user cost and should increase desired capital.

3.2 Financial constraints and "additionality"

A key mechanism is that the same subsidy can have different real effects depending on the firm's financing environment.

- **Perfect capital markets:** the supply of funds is flat at cost ρ, so the subsidy shifts down ρ and increases optimal capital.
- **Imperfect capital markets:** the marginal cost of funds is upward sloping (external finance is more expensive). A subsidy can relax the effective financing constraint, yielding a *larger* capital response for constrained firms.
- **Large firms:** if they are not constrained and/or can strategically obtain grants without changing investment, subsidies can be largely inframarginal rents (low "additionality").

Figure 2 in the paper illustrates these cases graphically.

3.3 Labor effects: substitution vs scale

Lower capital cost raises capital demand. Employment can rise or fall depending on:

- **Substitution effect:** capital becomes cheaper $\Rightarrow$ substitute capital for labor (reduces L).
- **Scale effect:** marginal cost falls $\Rightarrow$ output expands (raises L).

In a standard competitive environment with constant returns and product demand elasticity η and elasticity of substitution σ, the derived demand elasticity of labor with respect to the user cost of capital can be written as

$$\eta_{L,\rho} = s_K(\sigma - \eta), \tag{3}$$

where s_K is capital's share in total costs. Since $\partial\rho/\partial\phi < 0$,

$$\operatorname{sign}\left(\frac{\partial \ln L}{\partial \phi}\right) = \operatorname{sign}(-\eta_{L,\rho}) = \operatorname{sign}(\eta - \sigma)\,. \tag{4}$$

Interpretation:

- If demand is very elastic (η large), scale effects dominate and employment rises.
- If substitution is very strong (σ large), capital substitutes for labor and employment can fall.

Empirically, the paper finds employment rises in manufacturing, consistent with a strong scale effect in subsidized areas.

4 Empirical strategy: how they get causal effects

The central empirical challenge is that disadvantaged areas may be trending differently, and policy may be targeted endogenously. The paper builds an instrument from exogenous rule changes.

4.1 Outcome construction: differences relative to a base year

Let $y_{r,t}$ be an outcome in area (ward) r and year t (e.g., ln manufacturing employment or ln unemployment). They work with differences relative to a base year (1997) to remove time-invariant area heterogeneity:

$$\Delta y_{r,t} \equiv y_{r,t} - y_{r,1997}. \tag{5}$$

Similarly define $\Delta \mathrm{NGE}_{r,t} = \mathrm{NGE}_{r,t} - \mathrm{NGE}_{r,1997}$.

4.2 Core area-level second stage

A baseline second-stage specification is:

$$\Delta y_{r,t} = \lambda\, \Delta \mathrm{NGE}_{r,t} + \tau_t + X'_{r,1993}\pi_t + \Delta v_{r,t}, \tag{6}$$

where:

- τ_t are year fixed effects (common shocks),
- $X_{r,1993}$ are predetermined characteristics used in the EU eligibility rules (entered flexibly/with interactions over time),
- λ is the causal effect of subsidy intensity (the policy parameter of interest).

4.3 How eligibility maps are determined: an index + cutoffs

Eligibility is determined by a rule applied to area characteristics. Let $X_{r,t}$ be a vector of area characteristics (GDP per capita, population density, unemployment, etc.). They model a latent index

$$S^*_{r,t} = \theta'_t X_{r,t} + u_{r,t}, \tag{7}$$

and the observed subsidy tier (the NGE category) is an ordered mapping of $S^*_{r,t}$ crossing thresholds (cutoffs). They estimate θ_{1993} and θ_{2000} using *ordered probit* on cross-sectional eligibility/tier outcomes for 1993 and 2000 (Table 3).

4.4 The "policy-rule" instrument (shift in weights × baseline characteristics)

Holding baseline characteristics fixed, a change in the EU rule weights induces a predicted change in eligibility:

$$z_r \equiv (\widehat{\theta}_{2000} - \widehat{\theta}_{1993})' X_{r,1993}. \tag{8}$$

This object shifts NGE because the *rule changed*, not because the area contemporaneously changed.

The first stage is:

$$\Delta \mathrm{NGE}_{r,t} = \kappa\, z_r + \tau_t + X'_{r,1993}\psi_t + \Delta e_{r,t}. \tag{9}$$

Then they estimate λ by 2SLS using z_r as the instrument for $\Delta\mathrm{NGE}_{r,t}$.

4.5 Plant-level and firm-level versions

The same logic is implemented at the plant and firm level by mapping each plant/firm to the NGE in its location (or an exposure-weighted average if the unit spans multiple locations).

Plant-level employment (Table 10). For plant p in ward $r(p)$:

$$\Delta \ln L_{p,t} = \lambda^{(P)}\, \Delta\mathrm{NGE}_{r(p),t} + \tau_t + X'_{r(p),1993}\pi_t + \Delta\varepsilon_{p,t}, \tag{10}$$

estimated by 2SLS with the same instrument $z_{r(p)}$.

Firm-level outcomes (Table 11). For firm f with geographic exposure summarized by $\mathrm{NGE}_{f,t}$:

$$\Delta y_{f,t} = \lambda^{(F)}\, \Delta\mathrm{NGE}_{f,t} + \tau_t + X'_{f,1993}\pi_t + \Delta u_{f,t}. \tag{11}$$

4.6 Alternative treatment: actual subsidies received (TOT-style)

Eligibility changes move the cap NGE, but actual take-up is incomplete. So they also estimate models where treatment is realized subsidy payments, e.g.

$$\Delta y_{r,t} = \beta\,\Delta \ln(\text{RSA subsidy}_{r,t}) + \tau_t + X'_{r,1993}\pi_t + \Delta\xi_{r,t}, \tag{12}$$

instrumenting $\Delta \ln(\text{RSA subsidy}_{r,t})$ using the same policy-rule shifter. This pushes interpretation closer to a *treatment-on-the-treated* elasticity (Table 5).

4.7 Spillovers/displacement: why aggregation matters (intuition for Table 8)

A stylized way to think about displacement is:

$$\Delta y_{r,t} = \lambda\,\Delta\text{NGE}_{r,t} + \chi\,\Delta\overline{\text{NGE}}_{-r,t} + \text{controls} + \text{error}, \tag{13}$$

where $\Delta\overline{\text{NGE}}_{-r,t}$ is the average policy intensity in neighboring wards (or in the same local labor market).

- If $\chi < 0$, jobs are displaced from neighbors (zero-sum within a labor market).
- If $\chi \geq 0$, there are positive spillovers (e.g., input-output linkages, local demand multipliers).

Aggregating to a labor-market unit (the TTWA) partially nets out within-market displacement; thus Table 8 is a practical diagnostic for whether ward-level estimates are mostly "within-market reshuffling" or true net gains.

5 Reading the figures

5.1 Figure 1: map-based intuition for variation

Figure 1 visualizes the assisted areas (and tiers) pre-2000 vs post-2000. The takeaway is that:

- many areas switch *into* or *out of* eligibility,
- many areas remain eligible but switch *tiers* (cap changes),
- the reform creates a strong source of spatial variation that is plausibly exogenous once conditioned on predetermined rule variables.

5.2 Figure 2: conceptual mechanism (cost of capital and constraints)

Figure 2 is the theoretical picture:

- Panel A (perfect capital markets): subsidy shifts down the user cost ρ, raising desired capital.
- Panel B (imperfect capital markets): constrained firms face an upward-sloping supply of funds, so the same subsidy generates a larger capital response (greater additionality).

This motivates heterogeneity by firm size and financial constraints.

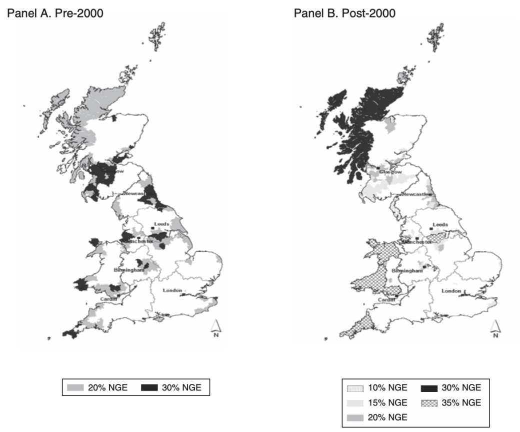

Figure 1. The Change in the Level of Maximum Investment Subsidy (NGE) between 1993 and 2000

Notes: The shaded areas are those that are eligible for some Regional Selective Assistance. In the 1993–1999 period, the dark shaded areas are the very deprived areas eligible for an investment subsidy of up to 30 percent NGE (the maximum investment subsidy, net grant equivalent). The light shaded areas are eligible for up to 20 percent NGE (see panel A). After 2000 Tier 1 areas had 35 percent NGE and Tier 2 (see panel B) areas ranged between 10 percent and 30 percent.
Source: Department of Business, Innovation and Skills "Industrial Development Reports," various years

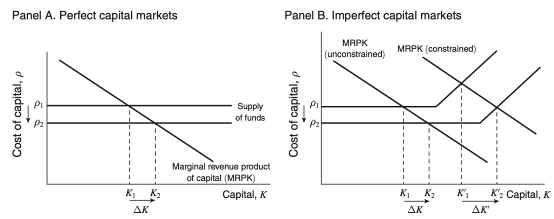

Figure 2. Effects of the RSA Policy on Capital

Notes: These figures examine the theoretical effect of the RSA policy reducing the cost of capital with perfect capital markets (panel A) and imperfect capital markets (panel B). For affected firms this is likely to raise capital, but the extent to which it does so will depend on a variety of factors such as whether a firm is financially constrained or more closely monitored (see text).

5.3 Figures 3 and 4: event-study style evidence

Figures 3–4 compare areas predicted to experience an *increase* versus *decrease* in support (as implied by the policy-rule instrument).

- Figure 3: manufacturing employment diverges after 2000 — areas predicted to gain support do better.
- Figure 4: unemployment falls relatively more in predicted-gain areas after 2000.

These graphs are a transparent "pre-trend check" and a visual complement to the IV regressions.

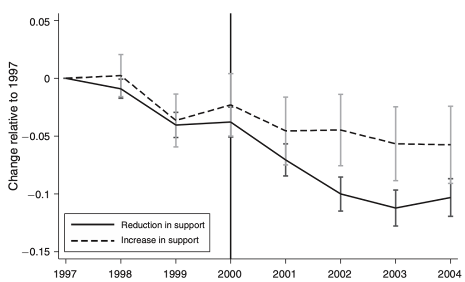

Figure 3. Changes in Manufacturing Employment in Areas with Increasing versus Decreasing Support Probability

Notes: Average changes relative to base year of 1997 in ln(employed) in a geographical area (ward). The dashed line shows average employment in wards that had an increase in support (as predicted by our policy rule IV). The solid line is average manufacturing employment in wards that had a decrease in support (as predicted by our policy rule IV). Ninety-five percent confidence bands also shown. The vertical line in 2000 shows when the change in policy occurred.

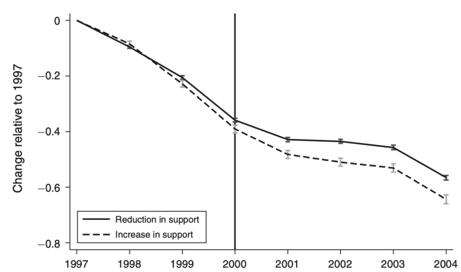

Figure 4. Changes in Unemployment in Areas with Increasing versus Decreasing Support Probability

Notes: Average changes relative to base year of 1997 in ln(number of unemployed) in a geographical area (ward). The dashed line shows average unemployment in wards that had an increase in support (as predicted by our policy rule IV). The solid line is average unemployment in wards that had a decrease in support (as predicted by our policy rule IV). Ninety-five percent confidence bands also shown. Unemployment is measured by those claiming unemployment insurance (job seekers allowance). The vertical line in 2000 shows when the change in policy occurred.

6 Interpreting every table

Below we interpret each table in the main paper in the order they appear.

6.1 Table 1: transitions in subsidy tiers (1993–1999 vs post-2000)

What it is: a transition matrix of wards across subsidy tiers before vs after the 2000 reform.
Why it matters: it shows large and nontrivial reclassification — the source of identification.
How to read: rows are pre-2000 tiers (0%, 20%, 30%); columns are post-2000 tiers (0%, 10%, 15%, 20%, 30%, 35%).
Key takeaway: many wards both (i) enter/exit eligibility and (ii) shift intensity conditional on being eligible.

TABLE 1—NUMBER OF AREAS (WARDS) ELIGIBLE FOR DIFFERENT MAXIMUM INVESTMENT SUBSIDIES (NGE) PRE AND POST POLICY CHANGE IN 2000

			Rate after 2000 (%)					
		Total	0	10	15	20	30	35
	Rate in 1993–1999 (%)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
1.	0	7,309	6,823	26	192	72	15	181
2.	20	2,012	841	102	539	33	118	379
3.	30	1,416	265	16	30	717	0	388
	Total	10,737	7,929	144	761	822	133	948

Notes: Table shows the numbers of wards in different regimes before and after the 2000 policy change. For example, it illustrates that of the 1,416 wards who were eligible for the maximum investment subsidy of 30 percent pre-2000 (column 1), 388 became eligible for the maximum subsidy of 35 percent after 2000 (column 7) and 265 lost their eligibility for subsidies completely (column 2).

TABLE 2—DESCRIPTIVE STATISTICS

	Years	Mean	Standard deviation	Observations	Number of units
Panel A. All areas					
NGE (maximum investment subsidy percent)	1997–1999	0.1	0.1	32,211	10,737
	2000–2004	0.1	0.1	53,685	10,737
Average RSA payment (£)	1997–1999	18,265.2	197,955	32,211	10,737
	2000–2004	9,837.1	134,036	53,685	10,737
Total unemployment (claimant count)	1997–1999	113.5	150.6	32,211	10,737
	2000–2004	80.9	112.5	53,685	10,737
Manufacturing employment	1997–1999	267.4	626.2	32,211	10,737
	2000–2004	233.0	535.9	53,685	10,737
Panel B. Areas eligible for RSA subsidies					
NGE (maximum investment subsidy percent)	1997–1999	0.2413	0.0492	10,284	3,428
	2000–2004	0.2367	0.08896	14,040	2,808
Average RSA payment (£)	1997–1999	56,132.2	346,827	10,284	3,428
	2000–2004	35,712.8	260,039	14,040	2,808
Total unemployment (claimant count)	1997–1999	161.85	179.04	10,284	3,428
	2000–2004	123.85	147.95	14,040	2,808
Manufacturing employment	1997–1999	350.96	823.91	10,284	3,428
	2000–2004	338.10	745.40	14,040	2,808
Panel C. Plant-level					
Plant employment across all areas	1997–1999	21.2	102.30	406,615	167,415
	2000–2004	19.8	91.90	631,089	183,061
Plant employment across eligible areas	1997–1999	27.4	139.00	131,431	53,575
	2000–2004	26.8	125.85	176,902	50,926
Panel D. Characteristics of recipients and non-recipients in eligible areas (£), firms					
Employment of recipients	1997–2004	87.2	323.4	16,413	4,550
Employment of non recipients	1997–2004	31.3	199.6	188,899	39,308
Investment of recipients	1997–2004	1,717.0	105,686	3,048	1,488
Investment of non recipients	1997–2004	953.3	8,001	15,314	7,449
(Value added/worker) recipients	1997–2004	38.6	25.8	3,048	1,488
(Value added/worker) non-recipients	1997–2004	44.9	231.51	15,314	7,449
TFP of recipients (indexed to industry × year average in logs)	1997–2004	–0.042	0.371	3,048	1,488
TFP of non recipients (indexed to industry × year average in logs)	1997–2004	–0.015	0.414	15,314	7,449

Note: TFP is computed using a Solow residual "factor share" method and relative to an industry × year average (see online Appendix C).

6.2 Table 2: descriptive statistics (areas, plants, and recipients)

What it is: means and standard deviations for:

- all wards and eligible wards (NGE, RSA payments, unemployment, manufacturing employment),
- plant-level employment (all vs eligible wards),
- firm characteristics comparing *recipients* vs *nonrecipients* in eligible areas.

Key descriptive patterns:

- Eligible areas have higher unemployment and higher manufacturing employment on average (they are disadvantaged/industrial areas).
- RSA payments decline over time in the sample.
- Recipients are substantially larger (higher employment, higher investment) than nonrecipients, suggesting strong selection into receipt and motivating IV.

6.3 Table 3: estimating the eligibility rule (ordered probit, 1993 vs 2000)

What it is: ordered probit estimates of how area characteristics map to subsidy tiers in 1993 and in 2000.

How it is used: the estimated coefficients $\widehat{\theta}_{1993}$ and $\widehat{\theta}_{2000}$ generate the policy-rule shifter $z_r = (\widehat{\theta}_{2000} - \widehat{\theta}_{1993})' X_{r,1993}$.

Interpretation of signs: richer / denser / more skilled areas are less likely to be eligible for high subsidies, while weaker labor-market conditions increase eligibility.

Key point: *the coefficients change between 1993 and 2000*, which is what creates the quasi-experimental "rule shock".

TABLE 3—ESTIMATES OF PARAMETERS ON ELIGIBILITY RULE CHANGES

	Main specification		Restricted variables	
	1993	2000	1993	2000
Dependent variable: level of NGE ordered variable	(1)	(2)	(3)	(4)
GDP per capita	−0.022	−0.040	−0.034	−0.055
	(0.002)	(0.002)	(0.002)	(0.002)
Population density	−0.028	−0.034	−0.043	−0.015
	(0.002)	(0.002)	(0.002)	(0.002)
Share of high skilled workers	−0.584	−1.438	−0.904	−2.268
	(0.129)	(0.149)	(0.125)	(0.142)
Business start-up rate	−2.414		−0.490	
	(0.240)		(0.183)	
Structural unemployment rate	83.251	32.681		
	(2.483)	(2.315)		
Activity rate	−1.147	−1.934	−1.235	−1.879
	(0.250)	(0.263)	(0.237)	(0.252)
Employment rate		−8.201		−11.259
		(0.462)		(0.444)
Current unemployment rate	−9.148	18.276	84.330	
(claimant count)	(3.240)	(3.565)	(1.846)	
ILO unemployment rate		−5.682		−0.122
		(0.824)		(0.760)
Long-duration unemployment rate	0.472		5.501	
	(1.216)		(1.163)	
Share of manufacturing workers		−1.122		−1.870
		(0.202)		(0.196)
Observations (wards)	10,737	10,737	10,737	10,737
Cut-off 10%	0.000	−9.503	−0.579	−14.697
	(0.220)	(0.420)	(0.210)	(0.377)
Cut-off 15%				
	1.202	−9.426	0.478	−14.629
Cut-off 20%	(0.221)	(0.420)	(0.210)	(0.377)
Cut-off 30%		−8.938		−14.201
		(0.419)		(0.375)
Cut-off 35%				
		−8.272		−13.600
log likelihood	−5,525.405	−6,879.521	−6,126.595	−7,325.151

Notes: Coefficients (robust standard errors) from ordered probits of NGE maximum support categories for 1993 and 2000. Dependent variable in columns 1 and 3 takes values of 1 to 3 depending on the level of NGE (0, 20, and 30 percent) and in columns 2 and 4 a value of 1 to 6 (0, 10, 15, 20, 30, and 35 percent). See online Appendix Table A2 for variable definitions. *Restricted variables* drops (collinear) structural unemployment in columns 3 and 4 and claimant unemployment in column 4.

TABLE 4—AREA-LEVEL REGRESSIONS: INSTRUMENTING MAXIMUM INVESTMENT SUBSIDIES (NGE) WITH RULE CHANGE

	OLS	Reduced-form	First-stage	IV
	(1)	(2)	(3)	(4)
Panel A. Dependent variable: ln(manufacturing employment)				
Maximum investment subsidy, NGE	0.124			0.953
	(0.070)			(0.260)
Policy rule instrument		0.839	0.881	
		(0.228)	(0.033)	
Panel B. Dependent variable: ln(unemployment)				
Maximum investment subsidy, NGE	−0.137			−0.414
	(0.024)			(0.078)
Policy rule instrument		−0.365	0.881	
		(0.069)	(0.033)	
Panel C. Dependent variable: ln(non-manufacturing employment)				
Maximum investment subsidy, NGE	0.006			0.177
	(0.044)			(0.161)
Policy rule instrument		0.156	0.881	
		(0.141)	(0.033)	
Number of areas (wards)	10,737	10,737	10,737	10,737
Observations	85,896	85,896	85,896	85,896

Notes: Standard errors (in parentheses below coefficients) are clustered at the area (ward) level. NGE (net grant equivalent) is the level of the maximum investment subsidy in the area. All columns include a full set of linear (lagged) characteristics used to define eligibility in 1993 ($X_{r,93}$). The time period is 1997–2004. *Policy rule instrument* is described in text. All variables are in differences relative to the base year of 1997.

6.4 Table 4: main area-level IV (NGE → jobs and unemployment)

Specification: Eq. (6) with ΔNGE instrumented by Eq. (9). Outcomes are in differences relative to 1997.

Panel A (manufacturing employment): IV coefficient is about 0.953 (s.e. 0.260).

Economic magnitude: a 10 percentage point increase in NGE implies $\approx 0.953 \times 0.10 \approx 0.095$ log points $\approx 9.5\%$ higher manufacturing employment.

Panel B (unemployment): IV coefficient is about -0.414 (s.e. 0.078), so higher subsidies reduce unemployment.

Panel C (non-manufacturing employment): small and statistically weak.

Why OLS differs: OLS effects are much smaller, consistent with endogenous targeting/selection (e.g., worse areas get higher caps, biasing OLS downward).

6.5 Table 5: using actual subsidies received (ln RSA payments)

What changes: treatment is ln(RSA subsidy) rather than NGE caps.
Interpretation: the IV coefficient is an elasticity: a 1% increase in subsidies is associated with about a 0.288% increase in manufacturing employment (Panel A, IV).
Takeaway: results remain consistent with the cap-based analysis; realized subsidies matter for jobs/unemployment, but endogeneity is severe in OLS.

Table 5—Area Level Regressions: Instrumenting Amount of Subsidy with Rule Change

Method	OLS (1)	Reduced-form (2)	First-stage (3)	IV (4)
Panel A. Dependent variable: ln(manufacturing employment)				
ln(RSA subsidy)	0.012			0.288
	(0.002)			(0.134)
Policy rule instrument		0.839	2.909	
		(0.228)	(1.140)	
Panel B. Dependent variable: ln(unemployment)				
ln(RSA subsidy)	−0.002			−0.125
	(0.001)			(0.053)
Policy rule instrument		−0.365	2.909	
		(0.069)	(1.140)	
Panel C. Dependent variable: ln(non-manufacturing employment)				
ln(RSA subsidy)	0.001			0.054
	(0.002)			(0.052)
Policy rule instrument		0.156	2.909	
		(0.141)	(1.140)	
Number of areas (wards)	10,737	10,737	10,737	10,737
Observations	85,896	85,896	85,896	85,896

Notes: Standard errors (in parentheses below coefficients) are clustered at the area (ward) level. RSA subsidy is the amount of subsidy (in thousands of pounds sterling) that an area receives on average per year; i.e., for every area and for the post- and pre-2000 period we sum the subsidy amount and divide by the number of years in the period. All columns include a full set of linear (lagged) characteristics used to define eligibility in 1993 ($X_{r,93}$). The time period is 1997–2004. Policy rule instrument is described in text. All variables are in differences relative to the base year of 1997.

Table 6—Area-Level Regressions Accounting for Structural Funds (SF)

	OLS (1)	Reduced-form (2)	First-stage NGE (3)	First-stage SF (4)	IV (5)
Panel A. Dependent variable: ln(manufacturing employment)					
Maximum investment subsidy, NGE	0.098				0.999
	(0.081)				(0.328)
Structural fund, SF	0.038				−0.029
	(0.037)				(0.079)
NGE IV		0.792	0.816	0.805	
		(0.224)	(0.034)	(0.067)	
Structural fund IV		0.094	0.124	1.029	
		(0.059)	(0.011)	(0.026)	
Panel B. Dependent variable: ln(unemployment)					
Maximum investment subsidy, NGE	−0.099				−0.409
	(0.027)				(0.098)
Structural fund, SF	−0.061				−0.050
	(0.012)				(0.027)
NGE IV		−0.374	0.816	0.805	
		(0.066)	(0.034)	(0.067)	
Structural fund IV		−0.103	0.124	1.029	
		(0.021)	(0.011)	(0.026)	
Number of areas (wards)	10,737	10,737	10,737	10,737	10,737
Observations	85,896	85,896	85,896	85,896	85,896

Notes: Standard errors (in parentheses below coefficients) are clustered at the area (ward) level. NGE (net grant equivalent) is the level of the maximum investment subsidy in the area. SF (structural funds) is a dummy variable equal to 1 if an area is eligible for SF support. NGE IV is the policy rule change instrument we introduced before. Structural funds IV is a rule change instrument that is computed in a similar way as NGE IV, except that rather than NGE eligibility we use SF eligibility (see text). All columns include a full set of linear (lagged) characteristics used to define eligibility in 1993. The time period is 1997–2004. All variables are in differences relative to the base year of 1997.

6.6 Table 6: accounting for EU structural funds

Concern: eligibility for RSA may covary with other EU regional policies (structural funds).
What table does: controls for structural funds and instruments both RSA intensity and structural funds with separate policy-rule shifters.
Takeaway: the RSA effect on manufacturing employment/unemployment remains similar to the baseline; structural funds have their own (smaller) effects.

6.7 Table 7: controlling for other regional policies

Purpose: show the estimated RSA effect is not driven by omitted correlated policies.
Reading: the reduced-form and IV coefficients remain stable as additional policy controls are added.
Takeaway: strengthens the causal interpretation of the rule-change instrument.

6.8 Table 8: higher geographic aggregation (TTWA)

Motivation: test for displacement/spillovers as in Eq. (13).
What they do: aggregate outcomes to travel-to-work areas (TTWA), closer to a local labor market definition.
Takeaway: effects remain positive for manufacturing employment and negative for unemployment, suggesting limited offsetting displacement across wards inside the same labor market.

Table 7—Controlling for Other Policies (In Employment Regressions)

	ln(manufacturing employment)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A. Reduced-form								
Policy rule IV	0.900	0.824	0.840	0.837	0.813	0.831	0.815	0.768
	(0.228)	(0.228)	(0.228)	(0.228)	(0.229)	(0.228)	(0.231)	(0.232)
Employment zones	−0.037						−0.024	−0.029
	(0.025)						(0.025)	(0.025)
Coalfield regeneration trust		−0.052					−0.057	−0.058
		(0.021)					(0.020)	(0.020)
Regional venture capital funds			0.031				0.023	0.026
			(0.018)				(0.018)	(0.018)
Enterprise grants				−0.003			−0.009	−0.012
				(0.015)			(0.016)	(0.016)
New deal for communities					−0.046		−0.057	−0.051
					(0.023)		(0.023)	(0.023)
Devolution to Wales and Scotland						−0.029	−0.041	−0.055
						(0.020)	(0.021)	(0.022)
Structural fund								0.134
								(0.063)
Panel B. IV								
NGE	1.090	0.943	0.954	0.972	0.914	0.931	0.966	0.895
	(0.277)	(0.262)	(0.260)	(0.266)	(0.259)	(0.255)	(0.274)	(0.319)
Employment zones	−0.074						−0.048	−0.053
	(0.028)						(0.027)	(0.028)
Coalfield regeneration trust		−0.03					−0.040	−0.040
		(0.022)					(0.021)	(0.021)
Regional venture capital funds			0.044				0.032	0.034
			(0.018)				(0.018)	(0.018)
Enterprise grants				0.033			0.018	0.019
				(0.019)			(0.018)	(0.018)
New deal for communities					−0.059		−0.080	−0.077
					(0.023)		(0.023)	(0.024)
Devolution to Wales and Scotland						−0.072	−0.075	−0.076
						(0.023)	(0.023)	(0.022)
Structural fund								0.055
								(0.075)
Number of areas	10,737	10,737	10,737	10,737	10,737	10,737	10,737	10,737
Observations	85,896	85,896	85,896	85,896	85,896	85,896	85,896	85,896

Notes: Standard errors (in parentheses below coefficients) are clustered at the area (ward) level. NGE (net grant equivalent) is the level of the maximum investment subsidy in the area. The time period is 1997–2004. NGE policy rule IV is described in text. Panel A has a specification identical to column 2 in Panel A of Table 4 except additional policy variables have been included (see text). Panel B has a specification identical to column 4 in Panel A of Table 4 except additional policy variables have been included (see text). All variables are in differences relative to the base year of 1997.

Table 8—Higher Level of Aggregation: Travel to Work Area (TTWA)

	OLS	Reduced-form	First-stage	IV
	(1)	(2)	(3)	(4)
Panel A. Dependent variable: ln(manufacturing employment)				
Maximum investment subsidy, NGE	0.538			1.006
	(0.114)			(0.319)
Policy rule instrument		1.053	1.047	
		(0.362)	(0.222)	
Panel B. Dependent variable: ln(unemployment)				
Maximum investment subsidy, NGE	−0.263			−0.803
	(0.062)			(0.265)
Policy rule instrument		−0.840	1.047	
		(0.249)	(0.222)	
Number of areas (TTWAs)	322	322	322	322
Observations	2,576	2,576	2,576	2,576

Notes: Standard errors (in parentheses below coefficients) are clustered at the TTWA. NGE (net grant equivalent) is the level of the employment weighted average maximum investment subsidy rate in the area. Standard errors below coefficients are clustered by area (TTWA level) in all columns. All columns include a full set of linear (lagged) characteristics used to define eligibility in 1993. The time period is 1997–2004. Policy rule instrument is described in text. All variables are in differences relative to the base year of 1997.

6.9 Table 9: extensive margin (number of manufacturing plants)

Question: are employment effects driven by entry/exit (more plants) rather than expansion of incumbents?
Result: coefficients on NGE (and on ln subsidies) are small and statistically weak for ln(# plants).
Interpretation: effects mostly operate through the *intensive margin* (existing firms/plants expanding employment), not a large net increase in the count of plants.

6.10 Table 10: plant-level employment and heterogeneity by firm size

Key result: IV effects are strong for plants belonging to *small firms* (firm employment < 50), and close to zero for plants in *large firms*.
Interpretation: subsidies create jobs where constraints/additionality are most binding (small firms), while large firms may capture rents without changing employment.

6.11 Table 11: firm-level outcomes (employment, investment, output, TFP)

Using firm-level data (including the ARD subsample):

- **Employment:** positive IV effects (Panel A and B).
- **Capital investment:** strongly positive IV effect (Panel C), consistent with the user-cost model.
- **Output:** positive IV effect (Panel D).

Table 9—Effects at the Extensive Margin? Number of Manufacturing Plants as an Outcome

	ln(number of manufacturing plants)				
	OLS (1)	Reduced-form (2)	First-stage (3)		IV (4)
Panel A. Baseline					
Maximum investment subsidy, NGE	−0.025				0.209
	(0.033)				(0.106)
Policy rule instrument		0.184	0.881		
		(0.094)	(0.033)		
Panel B. RSA subsidy levels					
ln(RSA subsidy)	0.003				0.063
	(0.001)				(0.039)
Policy rule instrument		0.184	2.909		
		(0.094)	(1.140)		
Number of areas (wards)	10,737	10,737	10,737		10,737
Observations	85,896	85,896	85,896		85,896
			First-stage		
			NGE	SF	
Panel C. Including structural funds in baseline					
Maximum investment subsidy, NGE	−0.062				0.097
	(0.036)				(0.135)
Structural fund	0.037				0.047
	(0.017)				(0.034)
NGE IV		0.117	0.816	0.805	
		(0.093)	(0.034)	(0.067)	
Structural fund IV		0.060	0.124	1.029	
		(0.026)	(0.011)	(0.026)	
Number of areas (wards)	10,737	10,737	10,737	10,737	
Observations	85,896	85,896	85,896	85,896	
Panel D. Travel to work area					
Maximum investment subsidy, NGE	−0.036				0.029
	(0.053)				(0.126)
Policy rule instrument		0.030	1.047		
		(0.132)	(0.222)		
Number of areas (TTWA)	322	322	322		322
Observations	2,576	2,576	2,576		2,576

Notes: NGE (net grant equivalent) is the level of the maximum investment subsidy in the area. The time period is 1997–2004. Policy rule IV is described in text. Panel A has a specification identical to panel A of Table 4 except the dependent variable is the ln(number of manufacturing plants in area + 1). Panel B corresponds to Table 5, panel C to Table 6, and panel D to Table 8. All variables are in differences relative to the base year of 1997.

Table 10—Plant-Level Employment Regressions, Splits by Firm Size

	ln(manufacturing employment)			
	OLS (1)	Reduced-form (2)	First-stage (3)	IV (4)
Panel A. Pooled across all plants, 653,385 observations on 96,768 plants; 9,975 wards				
Maximum investment subsidy, NGE	0.011			0.463
	(0.025)			(0.089)
Policy rule instrument		0.312	0.675	
		(0.058)	(0.040)	
Panel B. Small (plants in firm with under 50 employees), 594,356 observations on 87,728 plants; 9,883 wards				
Maximum investment subsidy, NGE	0.006			0.441
	(0.026)			(0.095)
Policy rule instrument		0.299	0.678	
		(0.063)	(0.040)	
Panel C. Large (plants in firms with over 50 employees), 59,025 observations on 9,036 plants; 3,708 wards				
Maximum investment subsidy, NGE	0.027			0.070
	(0.055)			(0.203)
Policy rule instrument		0.045	0.642	
		(0.130)	(0.050)	

Notes: Standard errors below coefficients are clustered by area (ward level) in all columns. NGE (net grant equivalent) is the level of the maximum investment subsidy in the area. All columns include a full set of linear (lagged) characteristics used to define eligibility in 1993. The time period is 1997–2004. Policy rule instrument is described in text. All variables are in differences relative to the base year of 1997.

- **TFP:** no significant effect (Panel E); the policy scales activity but does not raise measured productivity.

7 Overall conclusion and intuition

7.1 What the paper establishes

- Place-based investment subsidies can meaningfully increase manufacturing employment and reduce unemployment in disadvantaged areas.
- The strongest "real" responses come from small firms; large firms appear less additional.
- Subsidies raise investment and output but not TFP.
- The evidence is more consistent with "job preservation/creation" than with a transformational "big push" that permanently shifts the local equilibrium.

7.2 How we interpret the no-TFP result (TFP/misallocation lens)

A coherent intuition is:

- The subsidy primarily operates as a *factor-price wedge* (lower effective capital cost), pushing firms to expand and retain workers.
- It does not necessarily improve technology, management, or allocative efficiency; thus firm-level Solow residuals do not rise.

Table 11—Firm Level: Effects on Jobs, Investment, Output, and TFP. Instrumenting Maximum Investment Subsidy with Rule Change

	OLS (1)	Reduced-form (2)	First-stage (3)	IV (4)
Panel A. Dependent variable: ln(employment), full sample (449,514 observations, 91,546 firms)				
NGE	0.039			0.670
	(0.024)			(0.078)
Policy rule instrument		0.493	0.735	
		(0.057)	(0.011)	
Panel B. Dependent variable: ln(employment), ARD sub sample (45,511 observations, 21,389 firms)				
NGE	0.163			0.564
	(0.057)			(0.168)
Policy Rule Instrument		0.444	0.787	
		(0.132)	(0.032)	
Panel C. Dependent variable: ln(capital investment), ARD sub sample (45,511 observations, 21,389 firms)				
NGE	0.249			1.668
	(0.304)			(0.750)
Policy rule instrument		1.313	0.787	
		(0.588)	(0.032)	
Panel D. Dependent variable: ln(output), ARD sub sample (45,511 observations, 21,389 firms)				
NGE	0.031			0.399
	(0.065)			(0.182)
Policy rule instrument		0.314	0.787	
		(0.143)	(0.032)	
Panel E. Dependent variable: ln(TFP), ARD sub sample (45,511 observations, 21,389 firms)				
NGE	−0.034			−0.071
	(0.043)			(0.099)
Policy rule instrument		−0.056	0.787	
		(0.078)	(0.032)	

Notes: Standard errors below coefficients are clustered by area (ward level) in all columns. Policy rule instrument is described in text. The time period is 1997–2004. TFP is computed using a "factor share" method and relative to an industry × year average (see online Appendix C). All columns include a full set of linear (lagged) characteristics used to define eligibility in 1993. All variables are in differences relative to the base year of 1997.

- At the aggregate level, keeping less productive firms active can reduce measured productivity via misallocation (Hsieh–Klenow logic), even if welfare may rise due to lower unemployment.

8 Conclusion

8.1 Summary

This paper provides a rare *causal* micro-econometric evaluation of a large-scale industrial policy: the UK *Regional Selective Assistance* (RSA), a discretionary **capital investment subsidy** targeted to disadvantaged places. The core identification insight is that the policy environment changed because EU state-aid rules changed how areas were classified into subsidy tiers, generating plausibly exogenous variation in the *maximum allowed subsidy rate* (Net Grant Equivalent, NGE).

After correcting for the endogeneity of policy targeting, the paper concludes that RSA **meaningfully increased manufacturing employment and reduced unemployment** in eligible locations. These effects operate mainly through the **intensive margin** (expansion/preservation of activity among existing firms), are **much stronger for smaller firms** (and close to zero for large firms), show **little evidence of displacement** to nearby non-eligible areas or to non-manufacturing sectors, and do **not** increase measured **total factor productivity (TFP)**. The policy therefore looks more successful at creating/preserving jobs than at raising productivity, and any welfare assessment must trade off employment benefits against the efficiency costs of capital subsidies and the possibility of misallocation.

8.2 Key empirical takeaways

1. **Manufacturing employment rises strongly with subsidy intensity.** The baseline IV estimate in Table 4, Panel A is

$$\widehat{\lambda}_{IV} \approx 0.953 \quad (0.260).$$

Hence a 10 percentage point increase in the maximum subsidy rate (e.g. $\Delta\text{NGE} = 0.10$) implies

$$\Delta \ln L^{mfg} \approx 0.953 \times 0.10 = 0.0953 \quad \Rightarrow \quad \%\Delta L^{mfg} \approx \exp(0.0953) - 1 \approx 10.0\%.$$

2. **Unemployment falls.** Table 4, Panel B reports

$$\widehat{\lambda}_{IV} \approx -0.414 \quad (0.078),$$

so $\Delta\text{NGE} = 0.10$ implies

$$\Delta \ln U \approx -0.0414 \quad \Rightarrow \quad \%\Delta U \approx \exp(-0.0414) - 1 \approx -4.1\%.$$

3. **No robust effect on non-manufacturing employment.** Table 4, Panel C yields an IV estimate that is not statistically distinguishable from zero.

4. **Heterogeneity: strong effects for small firms, essentially zero for large firms.** The micro evidence shows that the job response is concentrated in smaller firms (consistent with higher "additionality"), while large firms receive subsidies but do not expand employment (consistent with inframarginal transfers, bargaining, or gaming).

5. **Mechanism: intensive margin dominates; entry effects are weak.** Area-level results indicate that employment increases occur mainly through expansions of incumbent activity rather than large increases in the number of plants.

6. **Spillovers/displacement appear limited in the margins tested.** The positive employment effect is not primarily explained by offsetting declines among nonparticipants, nearby non-eligible areas, or uncovered sectors.

7. **No TFP effect.** Firm-level analysis finds increased investment/output but no statistically meaningful increase in TFP; if anything, measured aggregate productivity can fall because subsidized firms are, on average, less productive (a misallocation-type channel).

8.3 Magnitudes and policy interpretation ("cost per job" and why it is not welfare)

The paper translates its IV estimates into a rough fiscal efficiency metric. A counterfactual that abolishes RSA rather than redrawing the map implies a loss of about **156,000 manufacturing jobs**. Using average RSA spending plus administrative costs and a deadweight cost of taxation, the implied annual cost is about **£288 million**, which corresponds to a "cost per job" of about

$$\text{Cost per job} \approx \frac{288 \text{ million}}{156{,}000} \approx \pounds 1{,}846 \quad \text{(about \$3,541 in 2010 prices).}$$

The paper stresses that this is **not** a welfare calculation: it omits classic capital-subsidy deadweight losses in product and factor markets, and it does not fully internalize the aggregate productivity consequences of keeping less productive firms operating. However, it also highlights that unemployment reduction can be a **first-order benefit** in depressed labor markets, so the overall assessment can still be positive.

8.4 Why the results make economic sense (intuition)

A coherent way to interpret the findings is:

1. **RSA is a capital-cost wedge.** By subsidizing investment, RSA lowers the effective user cost of capital. Firms expand investment, which can raise labor demand through a *scale effect* (even if capital and labor are substitutes technologically).

2. **Employment rises where slack is high.** In disadvantaged areas, labor market slack is larger, so expansions translate into lower unemployment rather than immediately bidding up wages.

3. **Small firms respond because the subsidy relaxes constraints and has higher additionality.** Smaller firms are more likely to face financing frictions and tighter internal constraints. A marginal reduction in the cost of expanding capacity can therefore generate real changes in investment and employment.

4. **Large firms exhibit low additionality.** For large firms, the marginal project may have gone ahead anyway, so subsidies become inframarginal rents; additionally, large firms may be better at navigating discretionary programs. This produces weak employment responses despite observed subsidy receipt.

5. **TFP does not rise because the policy is not a technology intervention.** RSA changes factor prices (and survival/scale), not the production frontier. Moreover, supporting relatively low-productivity incumbents can mechanically depress measured aggregate productivity through misallocation, even when the policy improves employment outcomes.

8.5 No "big push" (dynamic asymmetry) takeaway

The paper finds **no evidence** that RSA creates self-sustaining long-run equilibria where gains persist after support is removed. Areas losing subsidies experience job losses of similar magnitude to the gains in areas receiving more support. This pattern is consistent with a policy that *continuously props up* activity rather than permanently transforming local fundamentals.

8.6 Takeaway

RSA-style place-based investment subsidies causally raise manufacturing jobs and reduce unemployment (especially via small-firm intensive-margin expansion), but do not raise TFP and may lower measured productivity through misallocation, so the policy is best understood as an employment-support tool rather than a productivity-enhancing "big push".